

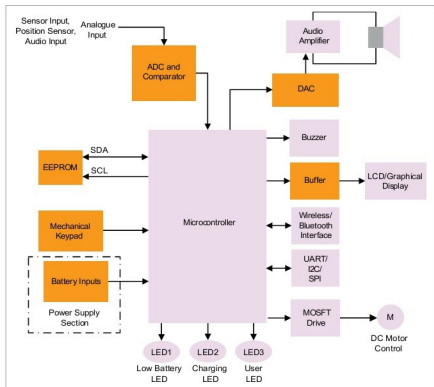


Fig. 5: Microcontrollers in toy applications

Application in electronic toys

Current electronic toys use 8/16-bit MCUs for motor control, LED/LCD/graphical display drives, different input and output interfaces, battery sensing and audio output. An MCU in chip-scale package (CSP) is most commonly used in electronic toys.

The MCU receives signals from analogue sensors (that receive analogue input and provide digital voltage, which is applied to the MCU) and 2/3-axis accelerometer (that measure 2/3-axis movement and convert it to digital voltage, which is applied to the MCU). It also uses the accelerometer and mechanical joystick for running gaming applications.

The MCU uses external buffer-driver circuitry to drive LED, LCD or graphical displays. Typically, LCD or graphical displays with backlight are used in low-end applications. The MCU also controls the backlight of LCD and graphical displays. It uses a touchscreen controller for the touchscreen interface

in high-end applications.

The MCU uses a mechanical keypad for inputs in low-end applications. It uses CapSense/touchscreen based keypads in high-end applications. The MCU interfaces with other onboard peripherals (like camera interface controller) through I2C, SPI and UART interfaces.

Toy applications use coin-cell or lithium-ion batteries for operation. The MCU monitors the coin-cell or lithium-ion battery voltage, and even battery-charging. It is also used for fault-detection and data-recovery applications.

The MCU used in proprietary 2.4GHz technology is targeted for remote-controlled toy. Gaming controllers and remote-controlled toys are also witnessing customers shifting from infrared to radio frequency.

System limitations and use of PSoC

A programmable system on chip (PSoC) is a combination of an MCU with programmable logic and high-

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